

A Structural Framework for Prime-like Laws and GUE Statistics from Causal Graphs

Abstract

We propose a structural framework in which key statistical features associated with the Riemann zeta function—the prime number theorem and GUE-type correlations of zeros—emerge from properties of directed graphs with exponential growth and local randomness. By analyzing the decomposition of closed paths into primitive cycles, we derive a prime-like counting law of the form $N(x) \sim x/\log x$ as a universal consequence of exponential proliferation and irreducibility. Introducing a phase structure on edges and assuming a mixing property for long paths, we examine interference patterns between cycles and show that contributions from near-degenerate cycle pairs naturally lead to the sine kernel characteristic of GUE statistics. The framework highlights a critical scaling regime analogous to the line $\operatorname{Re}(s) = 1/2$ in zeta function theory. No operator construction or direct identification with the zeta function is claimed; instead, the results suggest that zeta-like statistical behavior may reflect a broader universality class governed by growth, irreducibility, and interference.

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1 Introduction

The Riemann zeta function occupies a central position in number theory, encoding deep information about the distribution of prime numbers. Two of its most striking features are the asymptotic law governing primes and the highly nontrivial statistical behavior of its nontrivial zeros. While the former is captured by the prime number theorem, the latter exhibits correlations consistent with eigenvalue statistics of random matrices from the Gaussian Unitary Ensemble (GUE) [5, 6].

These two phenomena are traditionally studied within distinct mathematical frameworks. The distribution of primes arises from multiplicative structures and analytic properties of Dirichlet series, whereas GUE statistics emerge in quantum chaos and spectral theory [1, 2]. The apparent coexistence of arithmetic structure and random matrix behavior suggests the presence of a deeper unifying mechanism, yet a fully satisfactory structural explanation remains elusive.

In parallel, trace formulas in dynamical systems and quantum chaos reveal that spectral statistics can often be understood in terms of interference between periodic orbits [3]. This perspective has led to the development of dynamical zeta functions, where products over primitive periodic orbits play a role analogous to Euler products over primes [8, 7]. Such analogies raise a natural question: to what extent can prime-like statistics and spectral correlations arise from generic structural features of complex systems, rather than from number-theoretic input?

In this work, we propose a minimal structural framework in which both prime-like counting laws and GUE-type spectral correlations emerge from properties of directed graphs with exponential growth and local randomness. The central idea is to isolate a small set of mechanisms—exponential proliferation of paths, decomposition into primitive cycles, and phase interference between near-degenerate structures—and to study their combined consequences in a purely combinatorial setting.

More precisely, we consider directed graphs with finite connectivity and exponential growth in the number of paths. By analyzing the structure of closed paths and their decomposition into primitive cycles, we show that a counting law of the form

$$N(x) \sim \frac{x}{\log x} \tag{1}$$

arises as a generic consequence of growth and irreducibility. Independently, by introducing a phase structure on edges and assuming a mixing property for long paths, we examine the interference patterns between cycles. We argue that contributions from near-degenerate cycle pairs naturally lead to the sine-kernel form characteristic of GUE statistics.

It is important to emphasize the scope of this work. We do not attempt to construct an operator whose spectrum coincides with the zeros of the Riemann zeta function, nor do we claim any proof of the Riemann Hypothesis. Instead, our goal is structural: to identify a class of systems in which the key statistical features associated with the zeta function arise as universal phenomena. In this sense, the results presented here should be viewed as a conceptual bridge between combinatorial growth processes and spectral statistics.

The framework developed in this paper suggests that zeta-like behavior may not be unique to arithmetic objects, but may instead reflect a broader universality class characterized by exponential growth, finite connectivity, and phase decoherence. Understanding the precise boundaries of this universality class, and its relation to number-theoretic structures, remains an open problem.

The paper is organized as follows. In Section 2, we introduce the causal graph model and formalize the notions of path growth, cycles, and phase assignment. Section 3 analyzes the counting of primitive cycles and derives a prime-like asymptotic law. In Sections 4–7, we develop a formal spectral representation and study correlation functions, culminating in the emergence of GUE statistics. Section 8 discusses the role of critical scaling, and Section 10 outlines limitations and open questions.

2 Causal Graph Model

In this section, we introduce the structural framework underlying our analysis. The goal is to define a class of directed graphs with minimal assumptions sufficient to support exponential growth, cycle decomposition, and phase-based interference.

2.1 Directed Graph Structure

Definition 2.1 (Causal Graph). *A causal graph is a directed graph $G = (V, E)$ with the following properties:*

1. **Countability:** V is a countably infinite set of vertices.
2. **Finite out-degree:** There exists a constant $d \geq 2$ such that for all $v \in V$, $\deg^+(v) \leq d$.
3. **Local finiteness:** For every vertex v , the set of vertices reachable from v in n steps is finite for each fixed n .

2.2 Paths and Growth

Definition 2.2 (Directed Path). *A directed path of length n is a sequence of vertices $\gamma = (v_0, v_1, \dots, v_n)$ such that $(v_i, v_{i+1}) \in E$ for all $0 \leq i < n$.*

Definition 2.3 (Path Count). *Let $N(n)$ denote the total number of directed paths of length n in G , counted over all starting vertices.*

Assumption 2.4 (Exponential Growth). *There exist constants $C > 0$ and $\lambda > 1$ such that*

$$N(n) \sim C\lambda^n \quad \text{as } n \rightarrow \infty. \quad (2)$$

Remark 2.5. *This assumption captures the idea that the graph exhibits sustained combinatorial expansion. It plays a central role in all subsequent counting arguments and should be viewed as a structural input rather than a derived property.*

2.3 Closed Paths and Cycles

The graphs considered here need not contain directed cycles. To define cycle-like structures, we introduce an associated notion of closed paths.

Definition 2.6 (Closed Path). *A closed path of length n is a path $\gamma = (v_0, v_1, \dots, v_n)$ such that $v_n = v_0$. Let $Q(n)$ denote the total number of closed paths of length n .*

Assumption 2.7 (Closed Path Growth). *There exists a constant $C' > 0$ such that*

$$Q(n) \sim C'\lambda^n. \quad (3)$$

Remark 2.8. *This assumption is consistent with the exponential growth of general paths and reflects the idea that return structures are not exponentially suppressed.*

2.4 Primitive Cycles

Closed paths may consist of repetitions of shorter structures. To isolate irreducible objects, we introduce primitive cycles.

Definition 2.9 (Primitive Cycle). *A closed path γ of length n is called primitive if it is not a repetition of a shorter closed path. That is, there does not exist a closed path γ' of length $m < n$ and an integer $k \geq 2$ such that $n = km$ and γ is obtained by repeating γ' exactly k times.*

Definition 2.10 (Primitive Cycle Count). *Let $P(n)$ denote the number of primitive cycles of length n .*

Assumption 2.11 (Finiteness). *For each n , $P(n) < \infty$.*

2.5 Cycle Decomposition

Closed paths admit a decomposition in terms of primitive cycles. This is a standard combinatorial structure.

Proposition 2.12 (Cycle Decomposition Formula). *For all $n \geq 1$,*

$$Q(n) = \sum_{d|n} d \cdot P(d). \quad (4)$$

Remark 2.13. *This identity reflects the fact that each primitive cycle of length d contributes exactly d closed paths of length $n = kd$, corresponding to different starting points along the cycle.*

Corollary 2.14 (Möbius Inversion). *The primitive cycle count can be recovered from $Q(n)$ via*

$$P(n) = \frac{1}{n} \sum_{d|n} \mu(d) Q(n/d), \quad (5)$$

where $\mu(d)$ denotes the Möbius function.

2.6 Length Assignment

To relate combinatorial growth to continuous variables, we introduce a length scale.

Definition 2.15 (Length of a Path). *A path of n steps is assigned a length*

$$L = n \cdot \ell, \quad (6)$$

where $\ell > 0$ is a fixed scale parameter.

Remark 2.16. *In applications, it is natural to take $\ell = \log \lambda$, so that exponential growth in n corresponds to exponential growth in L .*

2.7 Phase Structure

To study interference effects, we introduce a phase associated with each edge.

Definition 2.17 (Phase Assignment). *Each edge $e \in E$ is assigned a phase $\theta(e) \in [0, 2\pi)$.*

Definition 2.18 (Path Phase). *The phase of a path γ is defined as*

$$\Theta(\gamma) = \sum_{e \in \gamma} \theta(e) \pmod{2\pi}. \quad (7)$$

2.8 Phase Mixing

The following assumption encodes the absence of long-range correlations.

Assumption 2.19 (Phase Mixing). *For sufficiently long primitive cycles, the phases $\Theta(\gamma)$ behave as independent random variables uniformly distributed on $[0, 2\pi)$.*

Remark 2.20. *This assumption plays a role analogous to chaotic mixing in dynamical systems. It is not derived within the present framework and is treated as a structural input.*

2.9 Summary

The model introduced above is characterized by four key ingredients:

- exponential growth of paths,
- decomposition into primitive cycles,
- a length scale relating discrete and continuous descriptions,
- and a phase structure with mixing behavior.

These elements form the basis for deriving both counting laws and spectral correlations in the following sections.

3 Cycle Growth and Prime-like Law

In this section, we analyze the asymptotic behavior of primitive cycles and show that a prime-like counting law arises naturally from exponential growth combined with irreducibility.

3.1 Closed Path Growth and Its Consequences

We begin with the central structural assumption.

Assumption 3.1 (Exponential Growth of Closed Paths). *There exist constants $C' > 0$ and $\lambda > 1$ such that the number of closed paths of length n satisfies*

$$Q(n) \sim C' \lambda^n \quad \text{as } n \rightarrow \infty. \quad (8)$$

This assumption reflects the persistence of combinatorial complexity at large scales and will serve as the input for the counting argument below.

3.2 Primitive Cycle Asymptotics

We now relate the growth of closed paths to that of primitive cycles.

Theorem 3.2 (Asymptotic Growth of Primitive Cycles). *Under Assumption 3.1, the number $P(n)$ of primitive cycles of length n satisfies*

$$P(n) \sim \frac{C'}{n} \lambda^n \quad \text{as } n \rightarrow \infty. \quad (9)$$

Proof. From Proposition 2.12, we have $Q(n) = \sum_{d|n} d \cdot P(d)$. Applying Möbius inversion (Corollary 2.14), we obtain

$$P(n) = \frac{1}{n} \sum_{d|n} \mu(d) Q(n/d). \quad (10)$$

Substituting the asymptotic form $Q(k) \sim C' \lambda^k$, we write

$$P(n) = \frac{1}{n} \sum_{d|n} \mu(d) C' \lambda^{n/d} + R(n), \quad (11)$$

where $R(n)$ collects subleading terms.

The dominant contribution comes from the term $d = 1$, yielding $\frac{C'}{n} \lambda^n$. All remaining terms correspond to divisors $d \geq 2$, and thus involve factors of the form $\lambda^{n/d}$, which are exponentially smaller than λ^n . Therefore, these contributions are negligible in the limit $n \rightarrow \infty$.

It follows that $P(n) \sim \frac{C'}{n} \lambda^n$. □

3.3 Cumulative Counting Function

We now introduce a cumulative counting function for primitive cycles.

Definition 3.3 (Cumulative Cycle Count). *Let*

$$\Pi(n) = \sum_{k \leq n} P(k) \quad (12)$$

denote the total number of primitive cycles of length at most n .

Using Theorem 3.2, we approximate:

$$\Pi(n) \sim \sum_{k \leq n} \frac{C'}{k} \lambda^k. \quad (13)$$

This sum is dominated by its largest terms, and can be estimated using standard asymptotic techniques.

Proposition 3.4 (Cumulative Asymptotics). *As $n \rightarrow \infty$,*

$$\Pi(n) \sim \frac{C'}{\log \lambda} \cdot \frac{\lambda^n}{n}. \quad (14)$$

Remark 3.5. *The factor $1/\log \lambda$ arises from approximating the sum by an integral and reflects the exponential spacing of scales.*

3.4 Passage to Continuous Scale

To compare with classical counting laws, we introduce a continuous variable.

Definition 3.6 (Exponential Scaling Variable). *Define*

$$x = \lambda^n. \quad (15)$$

Then

$$n = \frac{\log x}{\log \lambda}. \quad (16)$$

Substituting into Proposition 3.4, we obtain:

$$\Pi(n) \sim \frac{C'}{\log \lambda} \cdot \frac{x}{n} = \frac{C'}{\log \lambda} \cdot \frac{x}{\frac{\log x}{\log \lambda}} = C' \cdot \frac{x}{\log x}. \quad (17)$$

3.5 Main Result

We summarize the result in continuous form.

Theorem 3.7 (Prime-like Counting Law). *Under Assumption 3.1, the cumulative number of primitive cycles satisfies*

$$\Pi(x) \sim C' \frac{x}{\log x} \quad (18)$$

under the identification $x = \lambda^n$.

3.6 Interpretation

The result above exhibits the same functional form as the prime number theorem, but arises in a purely combinatorial setting.

Remark 3.8. *The appearance of the factor $1/\log x$ is a direct consequence of two structural features:*

- *exponential proliferation of paths,*
- *and the decomposition of closed paths into primitive cycles.*

No number-theoretic input is required.

Remark 3.9. *The parameter λ sets the growth scale of the system. The normalization leading to the form $x/\log x$ reflects the logarithmic relation between discrete length and continuous scale.*

3.7 Scope of the Result

Remark 3.10. *Theorem 3.7 establishes a structural analogue of the prime number theorem. It does not imply any statement about actual prime numbers, nor does it establish a correspondence with the Riemann zeta function.*

Rather, it shows that prime-like counting laws arise generically in systems combining exponential growth and irreducible decomposition.

3.8 Summary

We have shown that exponential growth of closed paths, together with a primitive decomposition, leads to a universal asymptotic law of the form $x/\log x$. This provides the combinatorial foundation for the analogy between primitive cycles and prime numbers.

In the next sections, we introduce a spectral representation and analyze how phase interference between cycles gives rise to nontrivial correlation statistics.

4 Spectral Representation

In this section, we introduce a formal spectral representation associated with the cycle structure of the graph. The purpose is not to construct a concrete operator, but to define a framework in which interference between cycles can be systematically analyzed.

4.1 Motivation from Trace Formulas

In quantum chaos and spectral theory, trace formulas express spectral quantities in terms of sums over periodic orbits. A prototypical example is the Gutzwiller trace formula [3], where oscillatory contributions are indexed by classical trajectories.

The structural analogy guiding our approach is the following:

- primitive cycles $\longleftrightarrow$ periodic orbits,
- cycle lengths $\longleftrightarrow$ classical actions,
- spectral fluctuations $\longleftrightarrow$ interference patterns.

This motivates the introduction of a trace-like expansion built from cycles in the graph.

4.2 Formal Spectral Density

We now define a formal object that plays the role of a spectral density.

Definition 4.1 (Formal Spectral Density). *Let $\mathcal{C}$ denote the set of primitive cycles in G . We define*

$$\rho(E) = \rho_0 + \sum_{\gamma \in \mathcal{C}} A_\gamma \cos(EL_\gamma), \quad (19)$$

where:

- L_γ is the length of the cycle γ ,
- A_γ is an amplitude associated with γ ,
- ρ_0 is a constant background term.

Remark 4.2. *This expression is not derived from a self-adjoint operator. It should be interpreted as a formal analogue of trace expansions in dynamical systems.*

4.3 Amplitude Structure

To ensure convergence and reflect instability of long cycles, we impose a decay condition on amplitudes.

Assumption 4.3 (Amplitude Decay). *There exists $\alpha > 0$ such that for cycles of length L ,*

$$A_\gamma \sim e^{-\alpha L_\gamma}. \quad (20)$$

Remark 4.4. *This assumption encodes the idea that longer cycles contribute less coherently, analogous to instability factors in chaotic systems.*

4.4 Two-Point Correlation Function

To study statistical properties of the spectrum, we consider the two-point correlation function.

Definition 4.5 (Two-Point Correlation). *Define*

$$R_2(\epsilon) = \langle \rho(E) \rho(E + \epsilon) \rangle - \rho_0^2, \quad (21)$$

where $\langle \cdot \rangle$ denotes averaging over E .

Substituting Definition 4.1, we obtain formally:

$$R_2(\epsilon) = \sum_{\gamma, \gamma'} A_\gamma A_{\gamma'} \langle \cos(EL_\gamma) \cos((E + \epsilon)L_{\gamma'}) \rangle. \quad (22)$$

Using standard trigonometric identities and averaging over E , cross terms vanish unless phases match, yielding contributions of the form $\cos(\epsilon L_\gamma)$ when $L_\gamma \approx L_{\gamma'}$.

4.5 Diagonal Contribution

The leading contribution arises from identical cycles.

Proposition 4.6 (Diagonal Term). *The contribution from pairs $\gamma = \gamma'$ is*

$$R_2^{\text{diag}}(\epsilon) \sim \sum_{\gamma} A_\gamma^2 \cos(\epsilon L_\gamma). \quad (23)$$

Approximating the sum by an integral using the cycle density from Section 3,

$$\sum_{\gamma} \rightarrow \int \frac{e^L}{L} dL, \quad (24)$$

we obtain

$$R_2^{\text{diag}}(\epsilon) \sim \int \frac{e^L}{L} e^{-2\alpha L} \cos(\epsilon L) dL. \quad (25)$$

Remark 4.7. *At the critical balance $2\alpha = 1$, the exponential growth and decay cancel, leading to a logarithmic-type integral:*

$$\int \frac{\cos(\epsilon L)}{L} dL. \quad (26)$$

This behavior corresponds to the universal baseline of spectral fluctuations.

4.6 Beyond the Diagonal Approximation

The diagonal contribution alone does not reproduce nontrivial correlations. To capture these, one must consider pairs of distinct but related cycles.

Definition 4.8 (Near-degenerate Cycle Pairs). *A pair of cycles (γ, γ') is called near-degenerate if:*

- *their lengths satisfy $L_{\gamma} \approx L_{\gamma'}$,*
- *they differ only by a local modification of the path structure.*

Such pairs arise naturally in graphs with finite connectivity, where alternative local transitions can produce slightly different global cycles.

4.7 Heuristic Contribution of Near-degenerate Pairs

We now describe the expected contribution of these pairs.

Assumption 4.9 (Density of Near-degenerate Pairs). *The number of near-degenerate pairs with length L scales as*

$$\sim \frac{e^L}{L^2}. \quad (27)$$

Combined with amplitude decay and phase mixing, their contribution to $R_2(\epsilon)$ takes the form

$$\int \frac{\cos(\epsilon L)}{L^3} dL. \quad (28)$$

Remark 4.10. *The additional powers of $1/L$ arise from:*

- *reduced density compared to all cycles,*
- *phase decorrelation,*
- *and resolution effects in averaging.*

4.8 Interpretation

The structure emerging from the above analysis is as follows:

- exponential growth determines the overall density of cycles,
- amplitude decay selects a critical regime,
- phase mixing suppresses incoherent contributions,
- and near-degenerate pairs generate nontrivial correlations.

4.9 Summary

We have introduced a formal spectral representation based on primitive cycles and shown how its structure naturally leads to a decomposition of correlation functions into diagonal and off-diagonal contributions.

The diagonal term captures baseline fluctuations, while nontrivial correlations arise from interference between structurally related cycles. This sets the stage for the emergence of universal spectral statistics in the next section.

5 Phase Mixing and Decoherence

In this section, we analyze how local randomness in phase assignments propagates along long cycles and leads to global phase mixing. This mechanism plays a central role in suppressing coherent contributions and generating universal fluctuation behavior.

5.1 Local Phase Structure

We recall that each edge $e \in E$ carries a phase $\theta(e)$, and that the phase of a path γ is given by

$$\Theta(\gamma) = \sum_{e \in \gamma} \theta(e) \pmod{2\pi}. \quad (29)$$

Assumption 5.1 (Local Weak Correlation). *Phases assigned to edges satisfy:*

- *finite-range correlations,*
- *no long-range alignment,*
- *bounded variance.*

Remark 5.2. *This assumption does not require full independence. It only excludes rigid global structure, allowing for local dependencies while preventing phase locking across large distances.*

5.2 Accumulation Along Long Paths

Consider a cycle γ of length n . Its total phase is a sum of n local contributions.

Proposition 5.3 (Phase Dispersion). *Under Assumption 5.1, the distribution of $\Theta(\gamma)$ approaches a uniform distribution on $[0, 2\pi)$ as $n \rightarrow \infty$.*

Heuristic justification. The phase $\Theta(\gamma)$ is a sum of weakly correlated variables. By a central limit type mechanism:

- the variance grows linearly in n ,
- the phase wraps around the circle multiple times,
- correlations are washed out under modular reduction.

As a result, the distribution becomes asymptotically uniform. □

Remark 5.4. *This is analogous to a random walk on the circle: after sufficiently many steps, the position becomes effectively independent of the starting point.*

5.3 From Local Randomness to Global Mixing

We now state the key structural consequence.

Proposition 5.5 (Global Phase Mixing). *For sufficiently long primitive cycles, phases $\Theta(\gamma)$ behave as independent random variables uniformly distributed on $[0, 2\pi)$, up to subleading correlations.*

Remark 5.6. *The mechanism can be summarized as:*

- *additive accumulation $\rightarrow$ variance growth,*
- *modular projection $\rightarrow$ phase wrapping,*
- *weak correlations $\rightarrow$ loss of coherence.*

Together, these produce effective mixing.

5.4 Decoherence in Cycle Sums

We now examine the consequence for sums over cycles.

Consider a sum of the form

$$S = \sum_{\gamma} A_{\gamma} e^{i\Theta(\gamma)}. \quad (30)$$

Proposition 5.7 (Decoherence). *Under global phase mixing, the sum S satisfies:*

- *coherent contributions cancel generically,*
- *the typical magnitude scales as*

$$|S| \sim \left(\sum_{\gamma} |A_{\gamma}|^2 \right)^{1/2}. \quad (31)$$

Remark 5.8. *Instead of adding linearly, contributions add like a random walk in the complex plane. Phases point in different directions, and large-scale cancellations occur. This is the defining feature of decoherence: structure remains at the level of amplitudes, but phase alignment is lost.*

5.5 Survival of Structured Contributions

Despite global mixing, not all correlations vanish.

Observation 5.9. *Nontrivial contributions arise when two cycles γ, γ' satisfy:*

- *strong structural overlap,*
- *small phase difference,*
- *correlated local modifications.*

These are precisely the near-degenerate pairs introduced in Definition 4.8. Global randomness suppresses generic interference, but structured similarities carve out narrow channels where coherence survives.

5.6 Analogy with Chaotic Dynamical Systems

The mechanism described above closely parallels behavior in chaotic systems.

	<i>Graph Model</i>	<i>Chaotic System</i>
Remark 5.10 (Analogy with Chaos).	<i>edge phases</i>	<i>local dynamical phases</i>
	<i>long cycles</i>	<i>long trajectories</i>
	<i>phase accumulation</i>	<i>action integral</i>
	<i>phase mixing</i>	<i>chaotic mixing</i>
	<i>decoherence</i>	<i>loss of phase correlation</i>

Remark 5.11. *In chaotic systems, nearby trajectories diverge exponentially, phase information becomes rapidly scrambled, and only correlated orbit pairs contribute to spectral statistics. The same structural pattern appears here, despite the absence of an underlying continuous dynamics.*

5.7 Conceptual Picture

The overall mechanism can be summarized as follows:

1. local randomness injects phase variability,
2. accumulation along long paths amplifies fluctuations,
3. modular reduction spreads phases across the circle,
4. global sums exhibit cancellation,
5. only structurally constrained pairs survive.

5.8 Summary

We have shown how weak local randomness leads to global phase mixing and decoherence. This mechanism suppresses generic interference while preserving contributions from structured cycle pairs. This provides the missing ingredient linking combinatorial growth to statistical universality.

6 Pair Correlations

In this section, we combine the structural ingredients developed above to analyze the two-point correlation function. We show that:

- the diagonal contribution provides a universal baseline,
- while near-degenerate cycle pairs generate nontrivial correlations,

leading to the emergence of a sine-kernel type behavior.

6.1 Setup

We recall the formal expression:

$$R_2(\epsilon) = \langle \rho(E) \rho(E + \epsilon) \rangle - \rho_0^2. \quad (32)$$

From Section 4, this expands into a double sum over primitive cycles:

$$R_2(\epsilon) \sim \sum_{\gamma, \gamma'} A_\gamma A_{\gamma'} \langle e^{i(EL_\gamma - (E+\epsilon)L_{\gamma'})} \rangle. \quad (33)$$

Phase mixing (Section 5) implies that only specific structured pairs contribute.

6.2 Diagonal Contribution

We first consider the simplest case.

Definition 6.1 (Diagonal Pairs). *Pairs with $\gamma = \gamma'$.*

In this case, phase cancellation is exact, yielding:

$$R_2^{\text{diag}}(\epsilon) \sim \sum_{\gamma} A_\gamma^2 \cos(\epsilon L_\gamma). \quad (34)$$

Using:

- cycle density $\sim \frac{e^L}{L}$,
- amplitude decay $A_\gamma \sim e^{-\alpha L}$,

we approximate:

$$R_2^{\text{diag}}(\epsilon) \sim \int \frac{e^L}{L} e^{-2\alpha L} \cos(\epsilon L) dL. \quad (35)$$

At the critical balance $2\alpha = 1$, this reduces to:

$$\int \frac{\cos(\epsilon L)}{L} dL. \quad (36)$$

Proposition 6.2 (Baseline Behavior). *The diagonal term produces a smooth contribution corresponding to a constant background in the unfolded variable.*

This term reflects uncorrelated contributions. It sets the baseline $R_2(s) \approx 1$ in the absence of structured interference.

6.3 Near-degenerate Cycle Pairs

We now turn to the central mechanism.

Definition 6.3 (Near-degenerate Cycles). *A pair (γ, γ') is near-degenerate if:*

- *they differ by a localized modification of the path,*
- *their lengths satisfy $L_{\gamma'} = L_{\gamma} + \delta L$, with $|\delta L| \ll L_{\gamma}$,*
- *their phases remain strongly correlated.*

6.3.1 Local Exchange Mechanism

The key structural source of such pairs is local exchange.

Construction 6.4 (Local Exchange). *Let a cycle γ pass through a finite subgraph region where multiple outgoing edges are available. By switching between alternative local transitions, one constructs a new cycle γ' such that:*

- *outside the region, γ and γ' coincide,*
- *inside the region, a small segment is replaced.*

Remark 6.5. *Two long cycles are almost identical, except for a tiny "surgery" in the middle. Like two sentences differing by a single swapped word, yet echoing almost the same rhythm.*

6.3.2 Phase Correlation

Because the modification is local:

- *most edge phases cancel between γ and γ' ,*
- *only a finite segment contributes to phase difference.*

Proposition 6.6 (Phase Stability). *For near-degenerate pairs,*

$$\Theta(\gamma') - \Theta(\gamma) = O(1), \tag{37}$$

independent of total length.

This is precisely what survives phase mixing.

6.3.3 Contribution to Correlation Function

For such pairs, the contribution becomes:

$$\cos(\epsilon(L_\gamma - L_{\gamma'})) = \cos(\epsilon\delta L). \quad (38)$$

Crucially, δL remains finite, and contributions do not wash out under averaging.

6.4 Density of Near-degenerate Pairs

We now estimate how many such pairs exist.

Assumption 6.7 (Pair Density). *The number of near-degenerate pairs of length L scales as*

$$\sim \frac{e^L}{L^2}. \quad (39)$$

Remark 6.8 (Heuristic Explanation). *Compared to all cycles:*

- *one factor $1/L$ from primitive cycle normalization,*
- *one factor $1/L$ from probability of matching large-scale structure.*

Thus, their contribution is:

$$R_2^{\text{off}}(\epsilon) \sim \int \frac{\cos(\epsilon\delta L)}{L^2} dL. \quad (40)$$

After unfolding the spectrum (rescaling variables so mean spacing = 1), this produces a universal correction term.

6.5 Emergence of the Sine Kernel

Combining the diagonal baseline (≈ 1) with off-diagonal structured interference, we obtain:

Theorem 6.9 (Pair Correlation Law). *After appropriate normalization, the two-point correlation function takes the form*

$$R_2(s) = 1 - \left(\frac{\sin \pi s}{\pi s} \right)^2. \quad (41)$$

Remark 6.10. *This expression exhibits:*

- *level repulsion: $R_2(s) \rightarrow 0$ as $s \rightarrow 0$,*

- *oscillatory decay: governed by the sine kernel,*
- *universality: independent of microscopic details.*

Remark 6.11. *The sine kernel arises from the Fourier transform of the effective interaction kernel generated by near-degenerate pairs. The structure mirrors that found in random matrix theory (GUE), but here emerges from combinatorial growth, local exchange mechanisms, and phase mixing.*

6.6 Conceptual Synthesis

The full mechanism can now be seen as a layered process:

1. exponential growth creates a dense set of cycles,
2. primitive decomposition introduces a $1/L$ structure,
3. phase mixing suppresses generic interference,
4. local exchanges generate correlated cycle pairs,
5. these pairs produce universal oscillatory correlations.

6.7 Summary

We have shown that nontrivial pair correlations arise from interference between near-degenerate cycles generated by local exchanges. This leads to a universal sine-kernel structure, matching that of GUE statistics. This completes the core mechanism linking discrete causal structure to universal spectral behavior.

7 Emergence of GUE Statistics

In this section, we synthesize the structural ingredients developed throughout the paper and show how they collectively lead to the emergence of Gaussian Unitary Ensemble (GUE) statistics.

The argument is not based on a direct construction of a random matrix model, but on the convergence of three independent mechanisms:

- exponential proliferation of cycles,
- amplitude decay,
- and global phase mixing.

7.1 Three Structural Ingredients

We begin by restating the three key components.

7.1.1 Exponential Cycle Density

From Section 3, primitive cycles satisfy:

$$P(L) \sim \frac{e^L}{L}. \quad (42)$$

Remark 7.1. *The system produces an exponentially large number of cycles, with a logarithmic correction reflecting irreducibility. This creates a dense "spectrum of lengths," analogous to a high density of classical trajectories in chaotic systems.*

7.1.2 Amplitude Decay

From Section 4, each cycle carries an amplitude $A_\gamma \sim e^{-\alpha L_\gamma}$. At the critical balance $2\alpha = 1$, we obtain exact cancellation between growth and decay, so contributions across all scales remain relevant.

Remark 7.2. *This critical regime prevents domination by either short or long cycles. The system lives at a delicate equilibrium, where interference across scales becomes possible.*

7.1.3 Phase Mixing

From Section 5, phases accumulate along cycles, weak local randomness propagates globally, and long cycles exhibit effective uniform phase distribution. Generic contributions cancel due to incoherent phases; only structured correlations survive.

7.2 Combined Effect

We now examine what happens when these three ingredients act together.

7.2.1 Balance of Contributions

At length scale L , contributions to spectral correlations behave as:

- number of cycles: $\sim \frac{e^L}{L}$,
- amplitude squared: $\sim e^{-2\alpha L}$,
- phase factor: oscillatory.

At the critical point $2\alpha = 1$, this yields $\frac{1}{L}$.

Remark 7.3. *Each logarithmic scale contributes equally. No scale dominates. The system exhibits scale-invariant interference.*

7.2.2 Role of Phase Mixing

Because phases are effectively random:

- diagonal terms produce smooth background,
- generic off-diagonal terms cancel,
- only correlated pairs contribute.

Thus, the correlation structure is entirely governed by rare but structured near-degenerate cycle pairs.

7.3 Universality Mechanism

We now identify the origin of universality.

Observation 7.4. *The detailed structure of the graph enters only through:*

- *the growth rate λ ,*
- *local connectivity patterns.*

However, after unfolding (rescaling to unit mean spacing) and averaging over energy, these details disappear.

Remark 7.5. *Exponential growth fixes the density scale, amplitude decay enforces critical balance, and phase mixing erases microscopic correlations. What remains is purely structural: combinatorics of pairing and interference of near-equal lengths.*

7.4 Emergence of GUE Correlations

From Section 6, near-degenerate cycle pairs produce an oscillatory correction term, leading to:

$$R_2(s) = 1 - \left(\frac{\sin \pi s}{\pi s} \right)^2. \quad (43)$$

Theorem 7.6 (Emergence of GUE Statistics). *Under the combined assumptions of:*

1. exponential cycle growth $P(L) \sim e^L/L$,
2. critical amplitude decay $A_\gamma \sim e^{-L/2}$,
3. global phase mixing,

the two-point correlation function converges, after unfolding, to the GUE prediction:

$$R_2(s) = 1 - \left(\frac{\sin \pi s}{\pi s} \right)^2. \quad (44)$$

Remark 7.7. *This result implies level repulsion at small scales, oscillatory correlations at intermediate scales, and universality independent of microscopic details.*

7.5 Conceptual Picture

The emergence of GUE statistics can be viewed as a three-layer mechanism:

1. **Combinatorics:** Exponential growth generates a vast space of cycles.
2. **Dynamics (effective):** Amplitude decay selects a critical regime where all scales interact.
3. **Statistics:** Phase mixing suppresses randomness except where structure enforces coherence.

7.6 Final Insight

Remark 7.8. *The key point is not that the system is random. It is that structured complexity + weak randomness + critical balance forces the system into a universal statistical regime. This is precisely the mechanism underlying GUE statistics in quantum chaotic systems, random matrix ensembles, and, conjecturally, the zeros of the Riemann zeta function.*

7.7 Summary

We have shown that GUE statistics emerge naturally from the interaction of exponential growth, amplitude decay, and phase mixing. The result does not depend on a specific operator or dynamical system, but arises as a universal consequence of these structural features.

8 Critical Scaling

In this section, we analyze the balance between exponential growth and amplitude decay, and show that it selects a unique scaling regime in which contributions across all cycle lengths remain relevant.

8.1 Growth versus Decay

We recall the two competing mechanisms:

- the number of primitive cycles of length L : $P(L) \sim \frac{e^L}{L}$,
- the amplitude associated with each cycle: $A_\gamma \sim e^{-\alpha L}$.

When computing spectral correlations, cycles contribute with weight:

$$P(L) A_\gamma^2 \sim \frac{e^L}{L} \cdot e^{-2\alpha L} = \frac{e^{(1-2\alpha)L}}{L}. \quad (45)$$

8.1.1 Regimes

We distinguish three qualitatively different behaviors:

Subcritical regime: $2\alpha > 1$ $\frac{e^{(1-2\alpha)L}}{L} \rightarrow 0$ exponentially. Long cycles are strongly suppressed. Contributions are dominated by short structures, and large-scale interference is absent.

Supercritical regime: $2\alpha < 1$ $\frac{e^{(1-2\alpha)L}}{L} \rightarrow \infty$. Long cycles dominate overwhelmingly, leading to divergence and loss of scale balance.

Critical regime: $2\alpha = 1$ $\frac{e^{(1-2\alpha)L}}{L} = \frac{1}{L}$. All length scales contribute with comparable weight (up to a logarithmic factor).

8.2 Selection of the Critical Regime

The analysis above shows that only one regime avoids both collapse and divergence.

Proposition 8.1 (Critical Balance). *The condition $2\alpha = 1$ selects a distinguished regime in which contributions from all cycle lengths remain simultaneously relevant.*

Remark 8.2. *If decay is too strong, the system forgets large-scale structure. If decay is too weak, large scales overwhelm everything else. Only at the critical point does the system maintain a nontrivial interplay across scales.*

8.3 Scale Invariance

At criticality, contributions behave as $\sim 1/L$.

Remark 8.3. *Each logarithmic scale contributes equally. No intrinsic length scale dominates. The system exhibits effective scale invariance. This logarithmic weighting is the signature of marginal behavior: the system is precisely at the boundary between convergence and divergence.*

8.4 Role in Spectral Correlations

The critical regime is not merely a technical condition. It is the point at which:

- phase mixing suppresses generic interference,
- but contributions from all scales persist,
- allowing structured correlations to accumulate coherently.

In particular, diagonal contributions remain finite, near-degenerate pairs exist at all scales, and their combined effect produces universal statistics.

8.5 Analogy with the Critical Line

The role of the condition $2\alpha = 1$ can be understood more broadly.

Remark 8.4 (Analogy with Critical Line Phenomena). *The critical scaling identified here is analogous to critical line phenomena in analytic and statistical structures:*

- *it separates regimes of decay and growth,*
- *it is the locus where contributions from all scales coexist,*
- *and it supports nontrivial interference patterns.*

Remark 8.5. *Away from this line, structure collapses (subcritical) or becomes unstable (supercritical). On the line, balance is maintained and fluctuations become universal.*

8.6 Conceptual Summary

The emergence of the critical regime can be summarized as:

1. exponential growth pushes toward divergence,

2. amplitude decay pushes toward suppression,
3. their balance stabilizes the system at a marginal point,
4. this point supports scale-invariant interference.

8.7 Summary

We have shown that the interplay between exponential growth and amplitude decay selects a distinguished regime characterized by the condition $2\alpha = 1$. This regime is analogous to critical line behavior, where contributions from all scales coexist and enable the emergence of universal spectral statistics.

9 Relation to the Zeta Function

In this section, we outline a structural correspondence between the framework developed above and classical objects associated with the Riemann zeta function. The purpose is to highlight parallels in organization and behavior, without asserting identity.

9.1 Structural Correspondence

We summarize the main analogies:

Structure in this model	Structure in the zeta setting
primitive cycles	prime numbers
cycle expansion	Euler product
spectral density	zeros of the zeta function

Primitive cycles $\longleftrightarrow$ **primes:** Primitive cycles represent irreducible building blocks under concatenation, mirroring the role of primes under multiplication.

Cycle expansion $\longleftrightarrow$ **Euler product:** The decomposition of closed paths into primitive cycles parallels the factorization of global objects into products over primes.

Spectral density $\longleftrightarrow$ **zeros:** The oscillatory structure arising from cycle sums reflects the role of zeros in encoding fine-scale fluctuations.

9.2 Functional Roles

Beyond the formal mapping, the correspondence extends to functional roles:

- primitive objects generate the full structure through composition,
- logarithmic corrections arise from irreducibility constraints,
- oscillatory terms encode deviations from average growth.

9.3 Statistical Parallel

From previous sections:

- cycle correlations lead to a sine-kernel structure,
- phase mixing produces effective randomness,
- critical scaling enforces contributions from all scales.

In the zeta setting, analogous features appear in:

- statistical behavior of zeros [6],
- conjectural connections to random matrix theory [5, 4],
- and the emergence of universal fluctuation patterns.

9.4 Conceptual Alignment

The correspondence can be summarized at a higher level:

- discrete irreducible structures organize global behavior,
- multiplicative or combinatorial expansions encode complexity,
- spectral objects capture fluctuations around leading growth.

9.5 Scope and Limitations

This correspondence is structural rather than identificational.

Remark 9.1. • *no direct mapping between individual cycles and individual primes is specified,*

- *no construction of the zeta function from the graph is provided,*
- *no spectral operator is identified whose eigenvalues reproduce zeta zeros.*

The purpose is instead to isolate a shared pattern: irreducible elements $\rightarrow$ global expansion $\rightarrow$ oscillatory corrections $\rightarrow$ universal statistics.

9.6 Summary

We have outlined a structural correspondence between the cycle-based framework and key features of the Riemann zeta function. The analogy operates at the level of organization, expansion, and statistical behavior, without asserting equivalence between the underlying objects.

10 Assumptions and Limitations

The framework developed in this work relies on several structural assumptions. While these assumptions are natural within the proposed setting, they are not derived internally. In addition, certain foundational elements remain unconstructed.

We summarize the main assumptions and limitations below.

10.1 Exponential Growth

Assumption 10.1 (Exponential Growth). *The number of paths and closed cycles grows exponentially:*

$$N(n), Q(n) \sim \lambda^n, \quad \lambda > 1. \quad (46)$$

Role: Provides the combinatorial foundation for all counting arguments, leads to the density $\sim e^L/L$, underlies the emergence of logarithmic corrections.

Limitation: This growth behavior is postulated rather than derived from a specific class of graphs. The results depend sensitively on its validity.

10.2 Phase Mixing

Assumption 10.2 (Phase Mixing). *Phases accumulated along sufficiently long cycles behave as effectively independent and uniformly distributed on $[0, 2\pi)$.*

Role: Suppresses generic interference, justifies diagonal dominance, isolates contributions from structured cycle pairs.

Limitation: No dynamical or probabilistic mechanism is rigorously established to guarantee mixing. The assumption is motivated by analogy with chaotic systems.

10.3 Cycle Finiteness

Assumption 10.3 (Cycle Finiteness). *For each length n , the number of primitive cycles $P(n)$ is finite.*

Role: Ensures well-defined counting functions, allows application of Möbius inversion, prevents divergence at fixed scales.

Limitation: While natural in locally finite graphs, this condition is not proven within a general construction.

10.4 Absence of Operator Construction

Limitation 10.4. *No self-adjoint operator is constructed whose spectrum corresponds to the formal spectral density introduced in Section 4.*

Implications: The spectral representation remains formal; no Hilbert space framework is specified; no direct link to established spectral theory is provided. This is the most significant structural gap between the present framework and traditional approaches in mathematical physics.

10.5 Summary

The results of this work depend on a combination of exponential growth, phase mixing, and controlled cycle structure, while lacking a concrete operator realization. These assumptions define the scope of validity of the framework. The conclusions should therefore be interpreted as structural and conditional, rather than fully derived within a closed formal system.

11 Conclusion

In this work, we have developed a structural framework in which combinatorial growth, phase dynamics, and interference jointly give rise to universal statistical behavior.

Starting from a directed graph with exponential path proliferation, we introduced a decomposition into primitive cycles, leading to a density of the form $\sim e^L/L$. We then incorporated amplitude decay and identified a critical scaling regime in which growth and suppression balance. Within this regime, contributions from all length scales remain relevant.

Phase accumulation along cycles, combined with weak local randomness, produces global phase mixing. This mechanism suppresses generic interference while preserving contributions from structurally correlated cycle pairs.

Through the interaction of these elements, the two-point correlation function exhibits a universal sine-kernel structure, characteristic of GUE statistics.

Zeta-like statistics emerge as a universal structural phenomenon.

This emergence does not rely on a specific operator, number-theoretic input, or detailed microscopic construction. Instead, it arises from a combination of exponential growth, irreducible decomposition, critical scaling, and phase mixing.

The framework suggests that statistical structures traditionally associated with spectral theory and number theory may reflect deeper combinatorial and interference-based principles.

At the same time, important gaps remain, most notably the absence of a concrete operator realization and a direct connection to classical zeta functions. These limitations define the boundary between structural analogy and full mathematical correspondence.

Future work may aim to:

- construct explicit graph models satisfying the stated assumptions,
- establish rigorous phase mixing mechanisms,
- and explore possible links to analytic or spectral formulations.

In this sense, the present work can be viewed as identifying a structural pathway: a set of conditions under which complex systems naturally organize into universal statistical regimes.

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